

EXPERIMENT 11

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1.

A company stores the monthly salary history of its employees in an EmployeeSalaryHistory table. Write an SQL query to find the employees whose salary has **increased in every recorded month compared to the immediately previous month**. For each salary record, compare the current month's salary with the employee's previous month's salary using a Window Function. Display only those employees for whom all later salary records are greater than their respective previous salary.

The output should display:

- Employee ID
- Employee Name

EmployeeSalaryHistory

record_id	employee_id	employee_name	salary_month	salary
1	101	Alok	2026-01-01	50000
2	101	Alok	2026-02-01	52000
3	101	Alok	2026-03-01	55000
4	102	Priya	2026-01-01	60000
5	102	Priya	2026-02-01	58000
6	102	Priya	2026-03-01	62000
7	103	Raj	2026-01-01	45000
8	103	Raj	2026-02-01	47000
9	103	Raj	2026-03-01	49000
10	104	Sneha	2026-01-01	70000
11	104	Sneha	2026-02-01	70000
12	104	Sneha	2026-03-01	72000

Output

employee_id	employee_name
101	Alok
103	Raj

Your query should select only those employees whose salary **increased in every recorded month compared to the immediately previous month**.

- **Alok (101):** Salary changes from 50000 → 52000 → 55000. Since the salary increases every month, Alok appears in the output.
- **Raj (103):** Salary changes from 45000 → 47000 → 49000. His salary also increases every month, so Raj appears in the output.
- **Priya (102)** is not included because her salary changes from 60000 → 58000 → 62000; it decreases once.
- **Sneha (104)** is not included because her salary changes from 70000 → 70000 → 72000. The salary remains the same once, whereas the requirement is a **strict increase (current salary > previous salary)**.

Therefore, only **Alok and Raj** satisfy the condition for all recorded months.

Solution:

```

SELECT EMPLOYEE_ID, EMPLOYEE_NAME
FROM (
  SELECT *,
    LAG(SALARY) OVER (
      PARTITION BY EMPLOYEE_ID
      ORDER BY SALARY_MONTH
    ) PREV_SALARY
  FROM EMPLOYEESALARYHISTORY
) E
GROUP BY EMPLOYEE_ID, EMPLOYEE_NAME
HAVING COUNT(*) FILTER (
  WHERE SALARY <= PREV_SALARY
) = 0;

```

2.

A company maintains employee details in an Employee table and department information in a Department table. Write an SQL query to find the employees whose salaries belong to the **top three distinct salary levels within their respective departments**. If multiple employees have the same salary, they should receive the same salary rank and all of them should be included in the result.

The output should display:

- Department Name
- Employee Name
- Salary

Department		Employee			
id	name	id	name	salary	departmentId
1	IT	1	Max	90000	1
2	Sales	2	Jim	85000	1
3	HR	3	John	85000	1
		4	Alex	75000	1
		5	Sam	70000	1
		6	Henry	80000	2
		7	Susan	75000	2
		8	David	70000	2
		9	Robert	65000	2
		10	Tina	60000	3
		11	Rita	55000	3
		12	Manoj	50000	3

Find employees
belonging to the **top
three distinct salaries of
each department.**

Output

Department	Employee	Salary
IT	Max	90000
IT	Jim	85000
IT	John	85000
IT	Alex	75000
Sales	Henry	80000
Sales	Susan	75000
Sales	David	70000
HR	Tina	60000
HR	Rita	55000
HR	Manoj	50000

Solution:

```

SELECT D.NAME AS DEPARTMENT,
E.NAME AS EMPLOYEE,
E.SALARY
FROM (
  SELECT *,
    DENSE_RANK() OVER (
      PARTITION BY DEPARTMENTID
      ORDER BY SALARY DESC
    ) AS RNK
  FROM EMPLOYEE
) E
JOIN DEPARTMENT D
ON E.DEPARTMENTID = D.ID
WHERE RNK <= 3;

```